MILITARY SPECIFICATION

FUZE FUNCTION CONTROL SET AN/AWW-l (*)

This notice forms a part of Specification MIL-F-21572(Wep) dated 15 July 1960 and shall constitute the covering page of the Specification

Specification MIL-F-21572(Wep) dated 15 July 1960 is hereby Downgraded from Confidential to UNCLASSIFIED

FSC 3390

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

WARNING: This material contains information affecting the national defense of the United States within the meaning of the Espionage Laws, Title 18, U.S. C., Section 793 and 794, the transmission or revelation of which in any manner to an unauthorized person is prohibited by law.

MIL-F-21572 (WEP)
15 JULY 1960
DECLASSIFIED
AUTH: MIL-F-21572 (WEP)
Notice 1 dtd 9 June 1964
T- INV. 00026 NSD PHILA., PA

MILITARY SPECIFICATION

FUZE FUNCTION CONTROL SET AN/AWW-l (*) (u)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be used in fighter-bomber type aircraft as a d-c and r-f fuze charging power-supply.

1.2 Classification - The Fuze Function Control Set AN/AlVW- 1 shall consist of the following items:

Items	Type Designation	Appl. Para.
Amplifier-Power Supply	AM-2014/AWW-1	3.4.1
Oscillator, Radio Frequency	O-562/AWW-1	3.4.2
Control, Fuze Function	C-2612/AWW-1	3.4.3

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental Aircraft ,Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro- Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection, For Aircraft Electronic Equipment, Format for

DECLASSIFIED
AUTH:MIL-F-21572 (WEP)
Notice 1 dtd 9 June 19??
T- INV. 00026 NSD PHILA., , F.A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200 Volt 3-Phase AC, or Combinations.

PUBLICATIONS

Bureau of Naval Weapons

SAR-300	Material Changes and Material Bulletins, Preparation of

DRAWINGS

Bureau of Naval Weapons

57A23ASK1	List of Drawings (BuAer) for Amplifier-Power Supply AM- 2014/AWW-l (also lists drawings for Oscillator, Radio Frequency O-562/AWW-1)
57A22ASK1	List of Drawings (BuAer) for Control, Fuze Function C-2612/AWW-l
57A2lD4	External Wiring Diagram (BuAer) for Fuze Function Control Set AN/AWW-l

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Copies of applicable drawings maybe obtained upon application to the Commanding Officer, U. S. Naval Avionics Facility, Indianapolis 18, Indiana. (See 6.3)

3. REQUIREMENTS

3.1. Parts and Materials- - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern

(1) Parts and materials requirements shall conform to Specification MIL-E- 5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the precuring activity. Non expendable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1. Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 16 7/8 pounds.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 3 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 1000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment snail have a minimum of 200 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS25245.

3.2.5 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

DECLASSIFIED
AUTH:MIL-F-21572 (WEP)
Notice 1 dtd 9 June 1964
T-INV. 00026 . NSD PHILA ., PA.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:54Z
Check the source to verify that this is the current version before use.

3.2.7 Interference Control- The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± l.O VAC and/or 27.5 ± 0.5 VDC

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental Service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.11.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E- 5400.

3.2.11.2 Temperature - The highest and lowest temperature conditions under which the equipment shall operate shall be + 55°C and -54°C respectively.

3.2.11.3 Cycle of Operation - Instead of continuous and intermittent operation, the equipment shall be operated according to the following applicable cycle.

(1) With high temperatures and normal ground altitude:

30 minutes on standby (tube heaters. energized)
15 minutes on full power (all channels operating)
15 minutes off (not considered as operating time)
Three cycles required

(2) With low temperatures and high-altitude pressures:

30 minutes on standby (tube heaters (energized)
5 minutes on full power (all channels operating)
5 minutes on standby
5 minutes on full power
5 minutes on standby
5 minutes on full power
Off

3.2.12 Warm-up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 15 minutes at extreme service conditions.

3.2.13 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-STD- 704 and sha11 give specified performance when provided with power from the following sources with characteristics as defined in MIL-STD-704. The power required shall not exceed the amounts shown:

AUTHORIZED 21572 (WEP)
(1) 115/200V 3Ø AC Category "C" 180.5 1 dtd 9 June 1964
T- INV. 00 6 NSD PHILA., PA

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:54Z
Check the source to verify that this is the current version before use.

(2) 28 V DC Category "C" 0.7 Amps.

3. 2.13.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits,

3.2.14 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.2.15 Continuity - The circuitry of the equipment shall conform to the applicable wiring diagrams.

3.2.16 Insulation Resistance - The electrical circuits of the equipment shall have an insulation resistance of at least five (5) megohms at 500 volts rms, 60 cycles per second, when measured between the housing and the isolated circuits, and between each and every isolated circuit.

3.2.17 Dielectric Strength - All electrical circuits in the equipment which are ungrounded and required to be insulated from each other shall be capable of withstanding 500 volts rms, 60 cycles per second, at room temperature and humidity for a period of ten (lO) seconds when this voltage is connected between each isolated circuit and the housing.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactor y performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.2 Standard Condition Performance Requirements - Under standard conditions (see 3.2.10), the equipment shall provide the adjusted and regulated outputs specified in Table I.

3.3.3 Service Condition Performance Requirements - Under service conditions (see 3.2.11), the equipment shal1 provide the adjusted and regulated outputs specified in Table L The tolerances given in the table shall be modified as follows:

(1) Each channel frequency shall remain within 1.6 percent of the center value given in the table.

(2) The regulated output voltage for each individual channel shall be 12 (+2, -3) volts mm.

(3) The regulated output voltages supplied by combinations of channels. shall be within ± 35 percent of the center value given in the table.

(4) All other voltage tolerances shall be as indicated in the table.

3.3.4 Thermal Delay Relay - The thermal delay relay K1208 shall operate within 30 ± 8 seconds after the application of 115 ± 10 volts, 400 ± 20 cycles, to pins m, f, and g of J1203.

3.3.5 Speed of Response - The release pulse circuit shall provide the voltages measured at pin B of J1202 within 20 milliseconds after + 28 volts d-c is applied to pin B of J1 204. The output voltage shall remain on pin B of J1202 approximately 1.0 second after removal of the + 28 volts d-c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

3.3.6 Amplitude Stability - At standard conditions, after a 2-minute warm-up period, and with the power supply on full power, each channel shall maintain its voltage amplitude within a tolerance of + 10 percent and -20 percent for 15 minutes.

3.3.7 R. F. Voltage - By adjusting the screen potentiometers, it shall be possible to raise the r-f voltage for each channel to 12 volts rms measured across a 60-ohm load in series with a l-microfarad capacitor at the output of the charging gear (pin B of J1202).

3.3.8 Output Waveform - Harmonic distortion shall not exceed 25 percent on the frequencies of channel numbers 5, 6, 7, and 8. Up to 40 percent distortion is acceptable on the other frequencies (channel numbers 1, 2, 3, and 4).

3.4 Detail Requirements -

3.4.1 Amplifier-Power Supply AM-2014/Aww- 1- The amplifier-power supply shall meet the following requirements:

3.4. 1.1 Function - The amplifier-power supply shall provide the voltages required by the oscillator circuits of Radio Frequency oscillator O-562/AWW- 1. It shall also supply the d-c and amplified r-f fuze charging outputs of the fuze function control set.

3.4.1.2 Form Factor - The dimensions of the amplifier-power supply shall not exceed 3 3/4 inches wide by 8 7/8 inches high by 11 7/16 inches long.

3.4.1.3 Weight - The weight of the amplifier-power supply shall not exceed 12 7/8 pounds.

3.4.1.4 Contents of Unit - The amplifier-power supply shall contain the following circuits.

(1) Power Supply Circuit - The power supply circuit shall provide the unregulated voltages required by the voltage regulator circuit and the cathode follower circuit.

(2) Voltage Regulator Circuit - The voltage regulator circuit shall provide regulated plate voltages for the amplifier circuit tubes (located in the amplifier-power supply) and the oscillator circuit tubes in Radio Frequent y Oscillator O- 562/AWW- 1. It shall also supply the regulated d-c fuze charging outputs of the fuze function control set.

(3) R-F Amplifier Circuit - The amplifier circuit shall amplify the r-f fuze charging signals provided by the Radio Frequency Oscillator O- 562/AwW- 1. The amplifier circuit shall contain a cathode follower circuit to provide a low impedance output with minimum r-f signal distortion.

3.4. 1.5 Controls - The front panel of the amplifier-power supply shall contain the following screwdriver adjusted controls.

(1) RF BIAS Control - The RF BIAS control shall be used to adjust the -55 ± O. 3 volt d-c grid bias on the cathode follower tubes. The control shall be labeled RF BIAS ADJ 55V.

(2) RF SUPPLY Control - The RF SUPPLY control shall be used. to adjust the +300 ± 1.5 volts d-c required by the amplifier circuit (in the amplifier-power supply) and the oscillator circuits of Radio Frequent y Oscillator O- 562/AWW- 1. The control shall be labeled RF SUPPLY ADJ 300V.

(3) OUTPUT Control (300V) - This OUTPUT control shall be used to adjust the 300 volt d-c fuze charging output of the fuze function control set. The control shall be labeled OUTPUT ADJ 300V.

(4) OUTPUT Control (195v) . This OUTPUT control shall be used to adjust the 195 volt d-c fuze charging output of the fuze function control set. The control shall be labeled OUTPUT ADJ 195 V.

DECLASSIFIED (WEP)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function,
J1201	58A5A189-1	RF signal inlet
J1202	58A6A51	Output receptacle
J1203	Ms3102A-28-21P	Power inlet and outlet to control panel
J1204	58A5A238-16-26SN	Power inlet and outlet to BULL PUP relay box

3.4.2 Oscillator, Radio Frequency O-562/AWW-l - The radio frequency oscillator shall meet the following requirements:

3.4.2.l Function- The radio frequency oscillator shall supply r-f fuze charging signals to the amplifier-power supply. The r-f signals, when amplified, shall meet the voltage and frequency requirements of table I.

3.4.2.2 Form Factor - The dimensions of the radio frequency oscillator shall not exceed 3 1/2 inches wide by 4 5/16 inches high by 4 5/8 inches long.

3.4.2.3 Weight - The weight of the radio frequency oscillator shall not exceed 1 3/4 pounds.

3.4.2.4 Contents of Unit - The radio frequent y oscillator shall contain four oscillator circuits with replaceable and interchangeable plug-in type tuned circuits. There shall be eight plug-in units which shall provide the frequencies specified in table I.

3.4.2.5 Controls - The radio frequency oscillator shall contain four screwdriver adjusted controls (one for oscillator circuit). The controls shall be used to adjust the amplitude of the r-f signal outputs. The controls shall be located on the side of the radio frequency oscillator and shall be labeled R1301 through R1304 respectively.

3.4.2.6 Electrical Connections - Connections to external circuits shall be provided as follows

Reference Designation	Receptacle Type	Function
JI3O1	58A5A189-2	DC inlet and RF outlet

3.4,3 Control, Fuze Function C-2612/AWW- 1- The fuze function control shall meet the applicable requirements of Specification MIL-C-6781 for Type I control panels.

3.4.3.1 Function- - The fuze function-control shall be used to select the available d-c and r-f fuze charging options.

3.4.3.2 Form Factor - The maximum dimensions of the fuze function control shall not exceed 5 3/4 inches wide by 3 inches high by 6 1/16 inches long.

3.4.3.3 Weight - The weight of the fuze function control shall not exceed 2.0 pounds.

3.4.3.4 Controls - The fuze function control shall contain the following operating controls on the front panel.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:54Z
Check the source to verify that this is the current version before use.

(1) SAFE - STBY - READY Switch - This 3-position rotary switch shall be used to control the power applied to the fuze function control set. The switch positions shall be labeled SAFE, STBY, and READY respectively.

(2) OPTION SELECTOR Switch - This 8-position rotary switch shall be used to select the various d-c and r-f fuze charging options available in the fuze function control-set. The switch shall be labeled OPTION SELECTOR, and the switch position shall be labeled A through H respectively.

(3) PRESS TO TEST Switch - This spring-loaded push button switch shall be used to check the system for readiness. The switch shall be labeled PRESS TO TEST.

3.4. 3.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1601	MS3102A- 28- 21P	Outlet to amplifier-power supply
J1602	58A5A238-14- 19PN	Power inlet and BULL PUP connection

3.4.3.6 Panel Illumination Lamps - When + 28 volts d-c is applied between pins p and b of J1 601, lamps 11601 and 11602 shall light.

3.4.3.7 HV D. C. CHECK Lamp - When+ 300 volts d-c is applied between pins K and b of J1601, and the PRESS TO TEST switch is pressed, lamp 11603 shall light.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(l) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractors Demonstration Tests - Contractor"s demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satifactory to the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:54Z
Check the source to verify that this is the current version before use.

4.2.2 Service Approval Tests -At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance. with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract . . . . . Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabri-. cation and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and 'that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material *or* performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, deliver y of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample -On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjetted to the individual tests. These tests shall, be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4. 1.1 Examination of Product - Each equinment shall be examined carefullv to determine that the material and workmanship requirements have been met.

4. 4. 1. 2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. .

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which-can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding *resonant* points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation "following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2- Sampling tests shall be conducted OR equiptments which have successfully. passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

4.4.2.1 Scope of Tests -As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the pre-production and special tests.

(3) Manufacturing reliability test specified in 4.4.1.3, except that the test duration shall be 30 hours.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the peformance of the equipment and to assure adequate quality control The equipment selected for special tests may be selected from equipments previously subjected to-the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production. equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken: .

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective activity intended to reduce the possibility of the future tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

(4) Where practical, include a test in the individual test to check all equipmerit for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test -A 300-hour test shall be conducted at the contractor's pkntonw early equipment representative of the current production The life test shall be performed under the conditions specified in 4.5.1. The, life test sample shall be selected by the government inspector in accordance -with the following. When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (O -5000 ft.)
Humidity	
A. C. voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check -At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4. 5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4. 5. 4. 1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples- An equipment which has been subjected to the life test shall reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items . The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary t o determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27. 5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, *or* applicable subsidiary specifications are in conflict, the following precedence shall apply:

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:54Z
Check the source to verify that this is the current version before use.

(1) Contract -The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Availability of Drawings - Complete sets of manufacturing drawings are available for the Fuze Function Control Set AN AWW- 1. These drawings are listed in List of Drawings 57A23AsKl and 57A22ASK1 (with the exception of Drawing 57A21D4). If drawings are required, request List of Drawings 57A23AsKl and 57A22ASKl and copies of all drawings listed therein, plus Drawing 57A21D4. Request for drawings must contain a statement of justification for actual need in relation to a government contract.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification NIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The parentheses (*), when used in the type designation, shall be replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:54Z
Check the source to verify that this is the current version before use.

Table I

FUZE FUNCTION CONTROL SET AN/AWW-l, ADJUSTMENTS AND OUTPUT REQUIREMENTS
(FUZE FUNCTION CONTROL C-2612/AWW-1 REMOVED)

INSTRUCTIONS:

(1) Ground pin J of J1203.
(2) Apply 115 volts, 400 cps to pins m, f, and g of J1203; if single phase is used, pins m, f, and g must be tied together,
(3) The voltage measured between pin A of J1204 and each of the pins a, b, and c of J1204 shall be 115 * 10 volts, 400 * 20 Cps.
(4) Apply +28 volts d-c to pins c and d (tied together) of J1203.
(5) Apply + 28 volts d-c to pin B of J1204.
(6) Adjust R1201 for r-f bias of -55.0 *O. 3 volts d-c.
(7) Adjust R1204 for r-f supply of 300.0 ± 1.5 volts d-c.
(8) Adjust R1203 for 300 ± 1.5 volts d-c. Measure between pin B of J1202 and ground with no load after a 15-minute warmup period.
(9) Adjust R1202 for 195 ± 1 volts d-c. Measure between pin B of J1202 and ground with no load after a 15-minute warmup period.

COLUMN HEADINGS

1 Template number
Oscillator plug-in socket
3 Channel number
4 Pins to be grounded on J1203
5 Resistor to be adjusted for r-f voltage output of 12 +0/-0.5 volts (rms); measured between pins A and B of J1202 across a 30 ohm load after a 15 minute warmup period
6 Regulated r-f output (volts rms) measured between pins A and B of J1202 across loads ranging from 30 to 120 ohms
7 Required frequency in kilocycles
8 D-C load (ohms) between pin B of J1202 and ground
9 D-C voltage (volts) measured between pin B of J1202 and ground

COLUMN HEADINGS ARE GIVEN ABOVE

1	2	3	4	5	6	7	8	9
I	XY1301	6	H	R1301	12 +1.4 -1.7	250,00 ± 1.25		
	XY1302	4	c	R1302	12 +1.4 -1.7	340,00 ± 1.70		
	XY1303	-						
	XY1304	8	E	R1304	12 +1.4 -1.7	184.00 ± 0.92		
	All of above	6,4,8	H,C, E	-	17.3	*5.2 -	10,000	300 ± 6
II	XY1301	1	H	R1301	12 +1.4 -1.7 W	539.00 ± 2.69		

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:54Z
Check the source to verify that this is the current version before use.

Table I [cont)

2	3	4	5	6	7	8	9
	XY1302	5	c	R1302	12 +1.4 -1.7	292.00 ± 1.46		
	XY1303	6	B	R1303	12 +1.4 -1.7	250.00 ± 1.25		
	XY1304	.						
	All of above	1,5,6	H,C, B		17.3 ± 5.2	-	10,000	300 ± 6
III	XY1301	1	H	R1301	12 +1.4 -1.7	539.00 ± 2.69		
	XY1302							
	XY1303	6 .	c	R1303	12 +1.4 -1.7	250.00 ± 1.25		
	XY1304							-
	All *Of above*	1,6	H,C		14.1	*4.2 -	10,000	**300 ± 6**
IV	XY1301	6	H	R1301	12 +1.4 -1.7	250.00 ± 1.25		
	XY1302	3	c	R1302	12 +1.4 -1.7	396.00 ± 1.98		
	XY1303	2	B	R1303	12 +1. 4 -1.7	462, 00 ± 2.31		
	XY1304							-
	All of above	6,3,2	H,C, B		17.3	+5.2 -	10,000”	**300 ± 6**
V	. . XY1301	6	H	R1301	12 +1.4 -1. 7	250.00 ± 1 25		
	XY1302	5- .	c	R1302	12 +1. 4 -1.7	292.00 ± 1.46		
	XY1303	-						
	XY1304	3	E	R1304	12 +1.4 -1.7	396.00 ± 1.98		
	All Of above	6,5,3	H,C, E		17.3 ± 5.2	-	10,000	300 * 6

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:54Z
Check the source to verify that this is the currentversion before use.

Table I (cont)

1	2	3	5	5	6	7	8	9
VI	XY1301	1	H	R1301	12 +1.4 -1.7	539. 00 *2.69		
	XY1302	7	c	R1302	12 +1. 4 -1.7	215. 00* 1.07		
	XY1303	6	B	R1303	12 +1.4 -1.7	250000 * 1.25		
	XY1304	-						
	All of above	1,7,6	H, C, B		17.3 ± 5.2 -		10,000	300 ± 6
VII	XY1301	5	H	R1301	12 +1. 4 -1.7	292.000 *1. 46		
	XY1302	7	c	R1302	12 +1. 4 -1.7	215. 00 *1.07		
	XY1303	1	B	R1303	12 +1.4 -1.7	539.00 ±2.69		
	XY1304	-						
	All Of above	5, ‘7,1	H, C, B		17.9 ± 5.2 -		10,000	300 ± 6
			h				10,000	-300 ± 6
			e	Adjust R1202 for 195 ± 1 volts d-c. Measure between pin B of J1202 and ground with no load after a 15-minute warmup period.				
			e	.			10,000	195 ± 4 ,
			e,h				10,000	-195 ± 4

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:54Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R066

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:54Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

PLATE NO. 15419 (BA

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:54Z
Check the source to verify that this is the current version before use.